

The herbal preparation STW 5 affects serotonergic pathways in the brain and colon as well as stress parameters in experimental irritable bowel syndrome

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Abstract

Background: Exposure to stress has been related to disturbance in 5-hydroxytryptamine (5-HT) signaling in the brain-gut axis and is considered as a major predisposing factor for the development of irritable bowel syndrome (IBS). The present study aimed to investigate the possible involvement of 5-HT and some other stress-related parameters in the effectiveness of STW 5 against stress-induced IBS.

Methods: Rats were subjected to restraint stress (RS) for 1 h/day for 14 consecutive days to induce IBS-like symptoms and were given STW 5 orally at the same time. At the end of the experiment, blood samples were withdrawn, then animals were euthanized and the brain hippocampi, cerebral cortices, as well as colons were isolated for biochemical and histopathological assessments.

Results: RS increased the plasma corticotrophin releasing factor (CRF) with concomitant increase in hippocampal and cortical 5-HT levels, as well as mast cell inflammatory mediators, oxidative stress biomarkers, and histopathological inflammatory changes observed in rat colon. It also decreased the colonic content of 5-HT with consequent decrease in fecal pellet output (FPO). Treatment with STW 5 protected against these changes.

Conclusion: The protective effect of STW 5 against RS-induced IBS is related to its ability to normalize the induced changes in 5-HT in the brain-gut axis and counteract the stress-induced oxidative stress and inflammation.

KEYWORDS

5-hydroxytryptamine, brain-gut axis, Restraint stress, STW 5

1 | INTRODUCTION

Irritable bowel syndrome (IBS) is a common functional gastrointestinal disorder characterized by recurrent abdominal pain and altered stool frequency or form Basnayake.¹ One of the predisposing factors related to IBS has been reported to be exposure to

stress. The clinical symptoms of IBS are often related to concomitant either diarrhea or constipation with probable involvement of 5-hydroxytryptamine (5-HT), since the latter has been shown to be a major excitatory neurotransmitter regulating intestinal movement and visceral nociception.^{2,3} Fichna and Storr⁴ reported that both plasma and colon 5-HT levels are elevated in diarrhea-predominant

IBS, but are reduced in constipation-predominant IBS. Most of the 5-HT in the gut is found in the enterochromaffin cells and in the serotonergic nerves of the myenteric plexus.⁵ The level of 5-HT in the gut has been shown to be related to activity along the brain-gut axis. The brain-gut axis (BGA) is a bidirectional communication network between the central nervous system (CNS) and gastrointestinal tract (GIT)⁶ and links the emotional and cognitive centers of the brain such as the cortex and hippocampus with the peripheral intestinal functions such as intestinal tract motility.⁷ One of the key signaling molecules within the BGA was reported to be 5-HT.⁸ Atkinson et al.⁹ and Sikander et al.,⁵ demonstrated that altered 5-HT signaling in BGA is closely related to development of functional GIT disorders such as IBS.

Exposure to stress is also known to activate the hypothalamic-pituitary-adrenal (HPA) axis leading to the release of the stress hormone corticotrophin releasing factor (CRF) from the hypothalamus. CRF, in turn, enhances the release of 5-HT and acts on a CRF receptor (CRF-R1) on the mucosal mast cells in GIT to enhance the release of mast cell mediators such as nerve growth factor (NGF), proteases, and cytokines (CKs). These mediators act on intestinal epithelium to increase intestinal permeability or to induce visceral hyperalgesia, representing two major pathophysiological features of IBS.^{10,11}

One of the herbal preparations that has been used to treat functional gastrointestinal disorders is STW 5, a combination of nine hydro-alcoholic herbal extracts.¹² STW 5 was shown to have modulatory effects on gastrointestinal motility and possesses anti-ulcerogenic, anti-oxidant, and anti-inflammatory properties.¹³ It has also been shown to be effective in a model of restraint stress-induced functional dyspepsia.¹⁴

Williams et al.,¹⁵ Kanauchi et al.,¹⁶ and Yan et al.,¹⁷ had previously shown that restraint stress could serve as an experimental model for IBS. Accordingly, this model was utilized in the present study with a view of exploring further the possible involvement of 5-HT, CRF, mast cell mediators, and oxidative stress biomarkers in the effectiveness of STW 5 in this condition.

2 | MATERIAL AND METHODS

2.1 | ANIMALS

Male albino Wistar rats, weighing 150–200 g each, were purchased from the Modern Veterinary Office for Laboratory Animals (Giza, Egypt) and housed 4 per cage, measuring 24 × 40 × 20 cm and containing sawdust as bedding, at a temperature of 25 ± 2°C, a relative humidity of 60%–70%, and a 12h/12h light/dark cycle (7 am to 7 pm). They were left to acclimatize for one week before subjecting them to experimentation. The study was approved by the Ethics Committee for Animal Experimentation at Faculty of Pharmacy, Cairo University (Permission Number: PT 2524), complying with the International Guiding Principles for Biomedical Research Involving Animals (1985).

2.2 | Drugs and Chemicals

STW 5 (Iberogast®) was generously provided by Steigerwald Arzneimittelwerk GmbH, Bayer Consumer Health (Darmstadt, Germany). It consists of a fixed proportion combination of hydro-alcoholic extracts (containing 31% alcohol) of nine herbs, namely, bitter candytuft (*Iberis amara*) (15%), peppermint leaves (*Mentha piperita*) (5%), chamomile flower (*Matricaria recutita*) (20%), liquorice root (*Glycyrrhiza glabra*) (10%), angelica root (*Angelica archangelica*) (10%), caraway fruit (*Carum carvi*) (10%), milk thistle fruit (*Silybum marianum*) (10%), lemon balm leaves (*Melissa officinalis*) (10%), and greater celandine herb (*Chelidonium majus*) (10%).

Rat specific ELISA kits for CRF (Cat. E1140Ra), NGF (Cat. E0539Ra), IL-1β (Cat. E0788Ra), and TNF-α (Cat. E0764Ra) were purchased from Bioassay Technology Laboratory (Shanghai, China) with sensitivities of 0.08 ng/ml, 5.01 ng/l, 10.23 pg/ml, and 2.51 pg/ml, respectively. 5-hydroxytryptamine hydrochloride was obtained from Sigma-Aldrich (St. Lois, Mo., USA). All other chemicals were of high analytical grade.

2.3 | Experimental design:

Rats were randomly allocated to five groups of 8 animals each. Two groups were kept as controls, while the other three were subjected to restraint stress (RS) by restraining them in tight well-ventilated cylindrical tubes for 1 h/day for 14 consecutive days.¹⁷ The animals were then subjected to experimentation as follows:

- (a) Group 1 (normal control): normal rats, given 1 ml/kg of 31% ethanol (vehicle for STW 5).
- (b) Group 2 (STW 5 control): normal rats treated with STW 5 (2.5 ml/kg).
- (c) Group 3 (RS control): RS rats, given 1 ml/kg of 31% ethanol.
- (d) Group 4: RS rats treated with STW 5 (1 ml/kg).
- (e) Group 5: RS rats treated with STW 5 (2.5 ml/kg).

All drugs were administered orally for the 14 consecutive days during which the animals were exposed to stress. All experiments were performed between 10–11 am to minimize the influence of changes in circadian rhythm. Twenty-four hours later, changes in body weight were assessed and the number of fecal pellets expelled by each rat in that hour was counted as an indirect measure of colonic transit.¹⁸ Pilot studies had shown that the FPO measured within the period of one hour, during which the animals were exposed to stress, was increased for the first 3 days then tended to revert to normal during the next 3 days. After that, the FPO was less than the normal output and remained constant until the end of the 2-week experimental period.

Blood samples were then withdrawn from the retro-orbital plexus of each rat under anesthesia (40 mg/kg ketamine/xylazine). Blood samples were collected into Eppendorf tubes containing heparin as an anti-coagulant and centrifuged at 825 g for 15 min at 4°C. The separated plasma was used to determine CRF levels using rat specific ELISA kit.

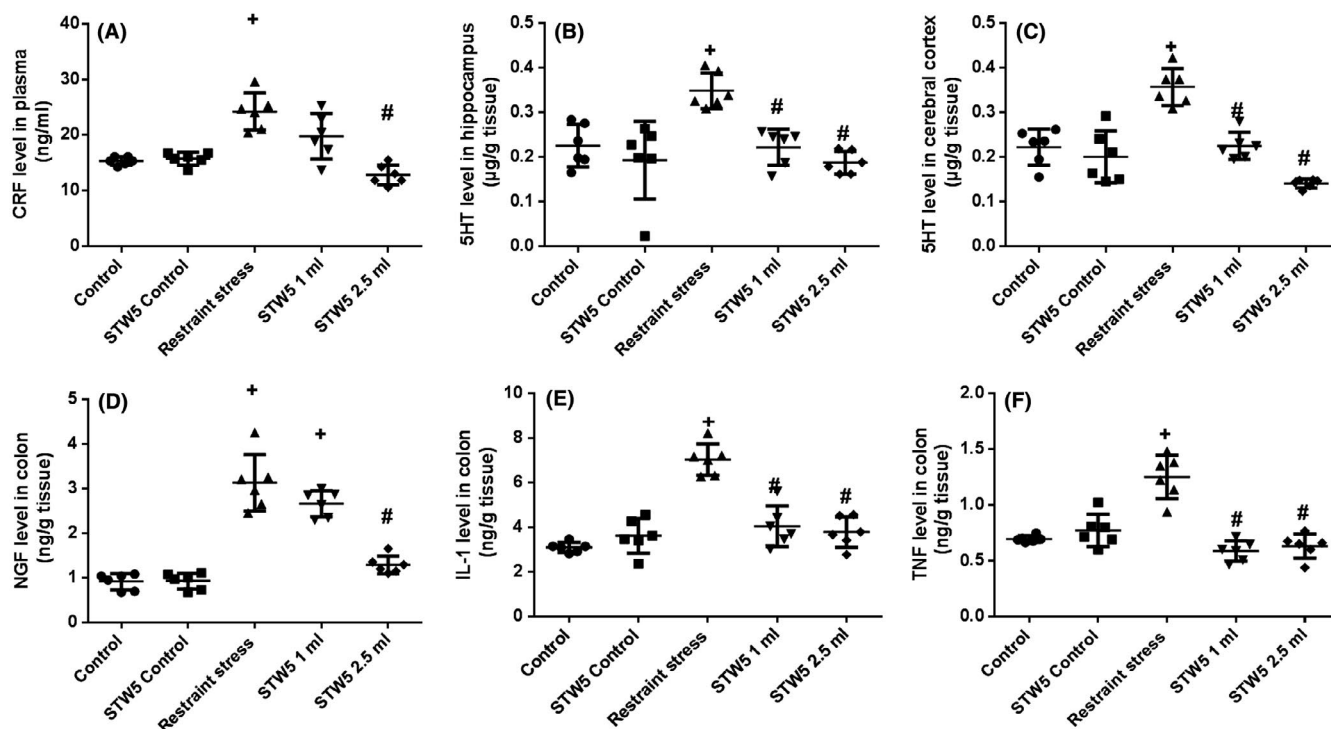


FIGURE 1 Effect of treatment with STW 5 on: (A) plasma level of corticotrophin releasing factor (CRF), (B) hippocampal content of 5-hydroxytryptamine (5-HT), (C) cerebro-cortical content of 5-HT, (D) colonic content of nerve growth factor (NGF), (E) colonic content of interleukin-1 β (IL-1 β), and (F) colonic content of tumor necrosis factor- α (TNF- α) in RS animals. Each whisker represents the mean $\pm$ SD ($n = 8$ rats). Statistical analysis was carried out by one-way ANOVA followed by Tukey multiple comparisons test; (*) significantly different from normal control group at $p \leq 0.05$; (#) significantly different from restraint stress group at $p \leq 0.05$

Animals were then sacrificed between 11 am–12 pm by decapitation (Guidelines of American Veterinary Medical Association, 2020 Edition) and the hippocampi and cerebral cortices were dissected out on dry ice. The distal 8 cm portion of the colon was also dissected out.

Each colon was divided longitudinally into two segments. The first segment was homogenized in normal saline to obtain a 10% homogenate (w/v) which was then centrifuged at 3300 g for 15 min at 4°C. The supernatant was used to measure the levels of the stress-related parameters NGF, IL-1 β , and TNF- α using respective ELISA kits. Moreover, glutathione (GSH), malondialdehyde (MDA), and nitric oxide (NO) were measured spectrophotometrically according to the methods described by Ellman¹⁹, Ohkawa et al.²⁰, and Montgomery and Dymock²¹, respectively.

The second segment of the colon was homogenized in 75% methanol to obtain a 10% homogenate (w/v) which was then centrifuged at 12,000 rpm for 20 min at 4°C and the supernatant used for estimation of 5-HT content by HPLC.²²

The determination of 5-HT in the hippocampi and cerebral cortices was carried out in the same way.

For histopathological examination, representative colon samples from each group were fixed in 10% formalin. Representative tissue sections (3 sections per animal), 5 μ m thin, were stained with hematoxylin & eosin (H&E) to examine histopathological changes, while others were stained with toluidine blue for the detection of mast cell infiltration.

2.4 | Statistical analysis

According to the Shapiro-Wilk test for normality, all the data of our findings were normally distributed except for the fecal output, which was therefore analyzed non-parametrically. Analysis of all the other data was carried out using one-way ANOVA followed by Tukey multiple comparisons test, while the FPO data was analyzed using Kruskal-Wallis followed by Dunn's *post hoc* test. The correlation between colonic 5-HT level and FPO values was analyzed using non-parametric Spearman's rank test, while the correlation between plasma CRF level and colonic NGF, IL-1 β , and TNF- α contents was carried out using parametric Pearson's rank test. For all statistical tests, the level of significance was set at $p \leq 0.05$. Statistical analysis was performed using GraphPad Prism 6 software (San Diego, USA).

3 | RESULTS

Exposure of the animals to RS increased the plasma CRF level by 60% with a nearly similar concomitant increase in the hippocampal and cortical 5-HT contents. Treatment with STW 5 at both dose levels protected against the changes in 5-HT in the two brain areas but only the higher dose could prevent the rise in plasma CRF (Figure 1A, B, and C).

The increase in plasma CRF was associated with an increase in the colonic contents of the mast cell mediators NGF, IL-1 β , and

TNF- α by 42%, 127%, and 80%, respectively. The changes in all these parameters were guarded against by STW 5 at its higher dose level (2.5 ml/kg), while the lower dose (1 ml/kg) was only effective in preventing the rise in IL-1 β and TNF- α (Figure 1D, E, and F).

Using Pearson's rank correlation test, a positive correlation was found between the plasma CRF level and colonic NGF ($r = 0.7954$), IL-1 β ($r = 0.6629$), and TNF- α ($r = 0.6065$) levels with P (two tailed) < 0.0001 (Figure 2A, B, and C).

RS reduced the colonic content of 5-HT to almost half and was associated with a decrease in FPO from about 5–6 in normal rats to nearly nil, indicating that RS exposure resulted in reduction in colonic motility and transit. These changes were prevented by concomitant treatment with STW 5 (Figure 3A and B). Moreover, there was a positive correlation between the colonic content of 5-HT and the FPO values using Spearman's rank test ($r = 0.5058$; $p = 0.0016$) (Figure 2D).

The oxidative stress biomarkers MDA and NO in the colon were also raised after exposure to RS by 33% and 56%, respectively, while the GSH content was reduced by 56%. Treatment of rats with STW 5 protected against these changes (Figure 4A, B, and C).

Concerning the changes in body weight, normal rats gained 30% in body weight over the 2-week experimental period as opposed to only 18% for rats subjected to RS. Treatment with STW 5 restored the normal rate of body weight gain (Figure 4D). The ANOVAs are summarized in table 1.

Histological examination of colon samples from both control untreated animals and control animals receiving only STW 5 showed a normal structure of the mucosa, submucosa, and muscularis (Figure 5). Exposure to RS led to focal inflammatory cell infiltration associated with congestion of blood vessels and edema in the submucosa, as well as lymphoid hyperplasia in the submucosa and mucosa. Staining with toluidine blue revealed mast cell infiltration in the submucosa of RS rats, but not in normal controls (Figure 6).

Treatment of RS rats with STW 5 (2.5 ml/kg) protected against the RS-induced histopathological alterations as shown by the normal histological structure of the mucosa, submucosa, and muscularis (Figure 7A and B). However, the lower dose of STW 5 (1 ml/kg) was less effective as sections still showed few focal inflammatory cell infiltrations with mild congestion in blood vessels and edema in the submucosa associated with few mast cell infiltrations in the submucosa (Figure 7C and D).

4 | DISCUSSION

5-hydroxytryptamine has been shown to be one of the key neurotransmitters in the brain neuroendocrine regions (hippocampus, hypothalamus, cortical regions, etc...) that is affected by stressors, the effect being partly mediated through the release of CRF.²³ Exposure of animals to chronic stress was indeed shown to activate the hypothalamic pituitary adrenal (HPA) axis resulting in the release of CRF and consequent increase in 5-HT content in the hippocampus.^{4,11} This is in accordance with the current results where RS exposure led to an increase in the plasma level of CRF with a concomitant rise in the hippocampal and cortical levels of 5-HT that is related to development of IBS-like symptoms. This in accordance with the work of Stasi et al.²⁴ who demonstrated that chronic stress affects cerebral areas (hippocampus, cortex, amygdala, and HPA axis) leading to local release of serotonin that results in hyperalgesia and allodynia typical of IBS. Earlier studies have also confirmed that restraint stress exposure leads to the development of IBS-like symptoms^{15-17,25}

The stress-induced rise in plasma CRF was shown to be positively correlated with an increase in the inflammatory mast cell mediators, NGF, IL-1 β , and TNF- α in the colon. This is in accordance with the work of Barreau et al.¹⁰ who demonstrated that CRF which is released after stress exposure acts on CRF-R1 receptor on mucosal mast cells leading to increased release of mast cell inflammatory

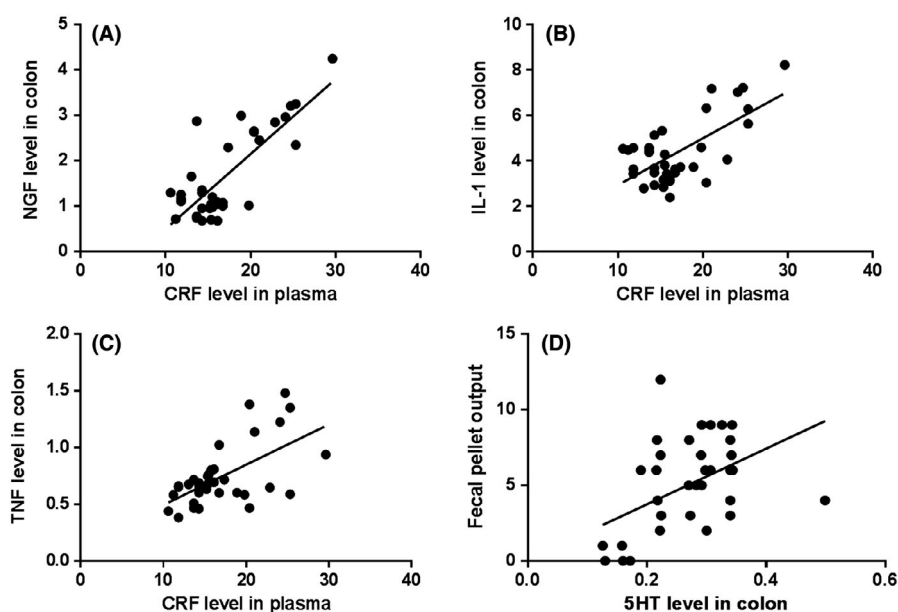


FIGURE 2 The correlation between plasma CRF level and the colonic (A) nerve growth factor (NGF), (B) interleukin-1 β (IL-1 β), and (C) tumor necrosis factor- α (TNF- α) contents. The correlation was carried out using Pearson's rank correlation test, while (D) the correlation between colonic 5-HT level and fecal pellet output values that was analyzed using Spearman's rank correlation test

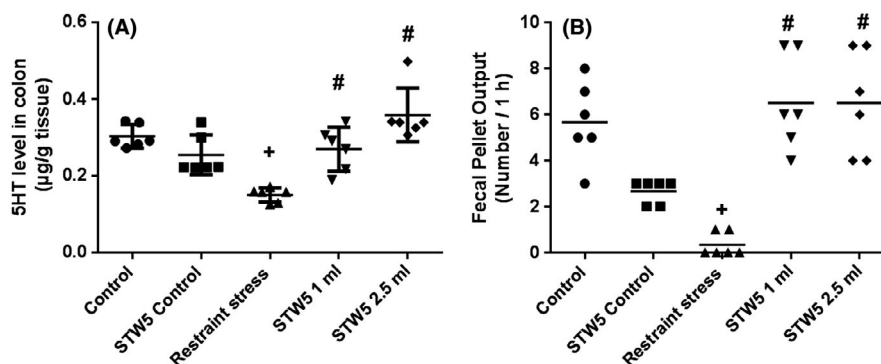
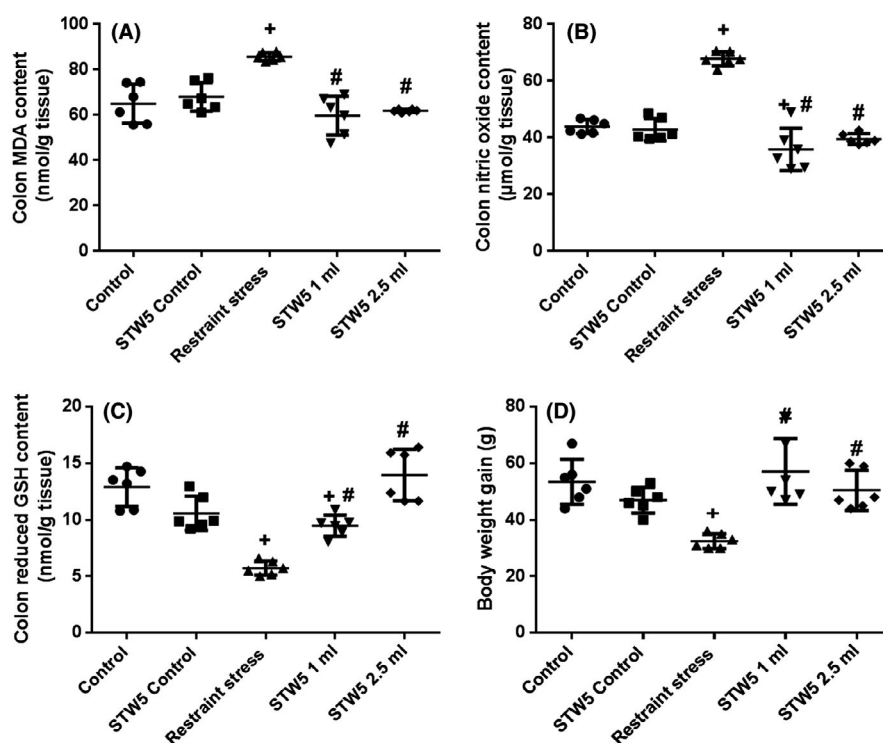


FIGURE 3 Effect of treatment with STW 5 on: (A) colonic content of 5-hydroxytryptamine (5-HT) in RS animals. Each whisker represents the mean $\pm$ SD (n = 8 rats). Statistical analysis was carried out by one-way ANOVA followed by Tukey multiple comparisons test. (B) fecal pellet output of RS animals. Each whisker represents the median with range (n = 8 rats). Statistical analysis was carried out by non-parametric one-way ANOVA (Kruskal-Wallis test) followed by Dunn's multiple comparisons test, (*) significantly different from normal control group at $p \leq 0.05$; (#) significantly different from restraint stress group at $p \leq 0.05$

FIGURE 4 Effect of treatment with STW 5 on: (A) colonic content of malondialdehyde (MDA), (B) colonic content of nitric oxide (NO), and (C) colonic content of reduced glutathione (GSH), (D) body weight gain in RS animals. Each whisker represents the mean $\pm$ SD (n = 8 rats). Statistical analysis was carried out by one-way ANOVA followed by Tukey multiple comparisons test; (*) significantly different from normal control group at $p \leq 0.05$; (#) significantly different from restraint stress group at $p \leq 0.05$



mediators such as NGF and cytokines like IL-1 β and TNF- α . These, in turn, were shown later to contribute toward the increased tight junction permeability, low grade inflammation, and visceral hypersensitivity associated with IBS.^{26,27}

Treatment of rats with STW 5 in a dose of 2.5 ml/kg protected against the rise in 5-HT level in both hippocampus and cortex as well against the rise in plasma CRF and consequent increase in colonic content of the mast cell mediators NGF, IL-1 β , and TNF- α . These results are in confirmation of previous studies by Abdel-Aziz et al.^{14,28} who demonstrated that exposure of rats to RS led to significant increase in plasma level of CRF and corticosterone in the esophagi of a rat model of reflux esophagitis with consequent rise in the level

of IL-1 β and TNF- α , while pretreatment with STW 5 effectively inhibited this rise. Moreover, Elbadawi et al.²⁹ reported that STW 5 possesses multiple anti-inflammatory as well as tight junction protective activities that could explain its clinically proven efficacy in managing IBS symptoms.

The present study also showed that repeated exposure to stress lowered the level of 5-HT in the colon. This was reflected in reduced colonic transit and motility and lowering of the fecal pellet output (FPO). Several previous studies had reported that whereas exposure to acute stress can increase colonic transit resulting in increased fecal pellet output,^{18,30,31} yet exposure to chronic restraint stress for 90 min for 5 consecutive days may initially increase the

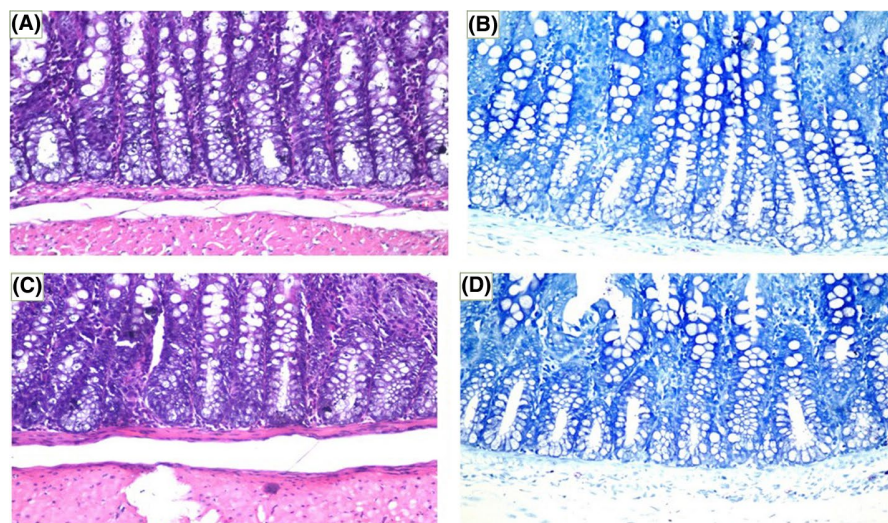


FIGURE 5 Photomicrographs showing: (A) histological structure of the colonic mucosa, submucosa, and muscularis of normal control rat (H&E, x40); (B) histological structure of the colon of normal control rat (toluidine blue, x40); (C) histological structure of the colonic mucosa, submucosa, and muscularis of control rat receiving STW 5 (2.5ml/kg) (H&E, x40); (D) histological structure of the colon of control rat receiving STW 5 (2.5ml/kg) (toluidine blue, x40)

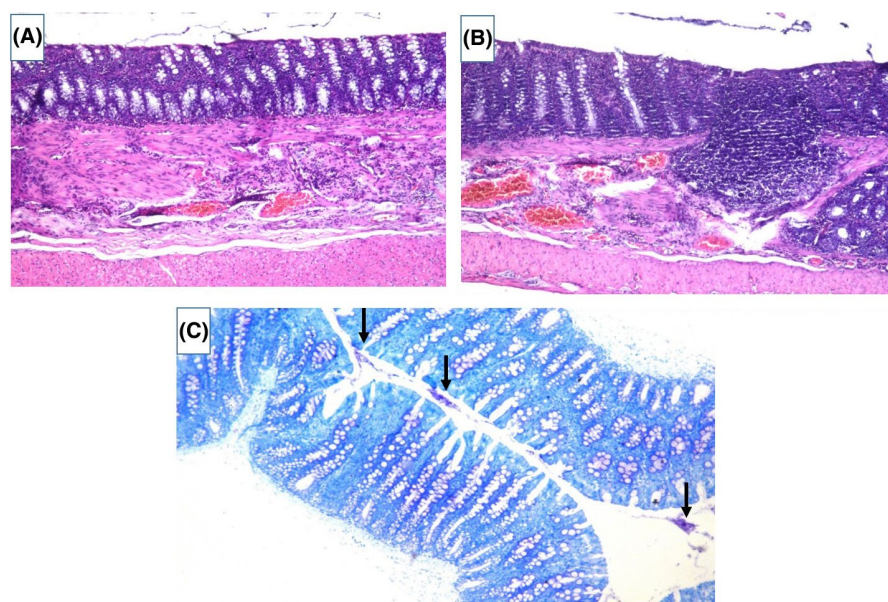


FIGURE 6 Photomicrographs of colon tissues of RS rats showing: (A) focal inflammatory cell infiltration associated with congestion in blood vessels and edema in the submucosa (H&E, x16); (B) lymphoid hyperplasia in the submucosa and extended to mucosa (H&E, x16); and (C) mast cell infiltration in the submucosal layer (toluidine blue, x16)

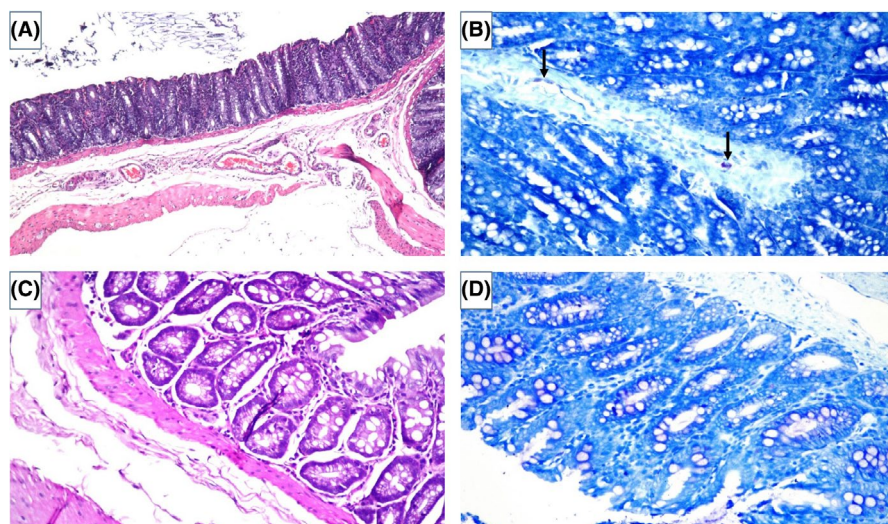


FIGURE 7 Photomicrographs of colon tissues of RS rats after treatment with STW 5 (1 ml/Kg) showing: (A) few focal inflammatory cell infiltration with mild congestion in blood vessels and edema (H&E, x16); (B) few mast cell infiltration in the submucosal layer (toluidine blue, x40), and after treatment with STW 5 (2.5 ml/Kg) showing: (C) normal histological structure of the colonic mucosa, submucosa, and muscularis (H&E, x40) (D) normal histological structure of the colon (toluidine blue, x40)

TABLE 1 Summary of one-way ANOVA results. Abbreviations: sum of squares (SS), mean squares (MS), degree of freedom (DF), 5-hydroxytryptamine (5-HT), nerve growth factor (NGF), interleukin-1 β (IL-1 β), tumor necrosis factor- α (TNF- α), malondialdehyde (MDA), nitric oxide (NO), and glutathione (GSH)

Parameter	Source	SS	DF	MS	F (DFn, DFd)	P-value
CRF	Between columns	475.2	4	118.8	F(4,25) = 17.97	<0.0001
	Within columns	6165.3	25	6.611		
	Total	40.4	29			
5-HT in hippocampus	Between columns	0.1039	4	0.02597	F(4,25) = 9.395	<0.0001
	Within columns	0.06912	25	0.002765		
	Total	0.1730	29			
5-HT in cortex	Between columns	0.1508	4	0.03770	F (4, 25) = 24.34	<0.0001
	Within columns	0.03872	25	0.001549		
	Total	0.1895	29			
NGF	Between columns	25.95	4	6.488	F (4, 25) = 54.93	<0.0001
	Within columns	2.953	25	0.1181		
	Total	28.90	29			
IL-1 β	Between columns	58.27	4	14.57	F (4, 25) = 29.86	<0.0001
	Within columns	12.20	25	0.4878		
	Total	70.47	29			
TNF- α	Between columns	1.733	4	0.4333	F (4, 25) = 26.97	<0.0001
	Within columns	0.4017	25	0.01607		
	Total	2.135	29			
5-HT in colon	Between columns	0.1410	4	0.03524	F (4, 25) = 14.60	<0.0001
	Within columns	0.06033	25	0.002413		
	Total	0.2013	29			
MDA	Between columns	2582	4	645.4	F (4, 25) = 16.93	<0.0001
	Within columns	953.2	25	38.13		
	Total	3535	29			
NO	Between columns	3832	4	958	F (4, 25) = 55.03	< 0.0001
	Within columns	435.2	25	17.41		
	Total	4267	29			
GSH	Between columns	250.2	4	62.56	F (4, 25) = 26.76	<0.0001
	Within columns	58.43	25	2.337		
	Total	308.7	29			
Body weight	Between columns	2170	4	542.5	F (4, 25) = 9.777	<0.0001
	Within columns	1387	25	55.49		
	Total	3557	29			

colon motility for the first 3 days, but reverts back to normal by the 5th day, suggesting that adaptation mechanisms develop following chronic stress³²

Treatment with STW 5 during the exposure to stress was shown to normalize the level of 5-HT in the colon as well as colonic transit. This was reflected by a tendency to normalize the FPO, an effect that contributes toward its therapeutic efficacy in treating functional GI disorders. The efficacy of STW 5 in normalizing the stomach and intestinal motility and reducing inflammation was recently reviewed by Kim et al.³³

Moreover, it was found that chronic stress exposure raised oxidative stress parameters in the colon tissue, evidenced by an elevation in MDA and NO contents and a reduction of GSH. Similar findings were reported by Colón et al.³⁴ who showed that RS increased the susceptibility of the rat colon to oxidative damage and increased nitric oxide synthase-2 activity. Treatment with STW 5 protected against these changes, possibly through its strong radical quenching activities³⁵ and anti-oxidant properties.³⁶

Exposure to RS was also reflected in histopathological changes in the colon. RS control rats showed increased inflammatory cell infiltration, blood vessel congestion, edema, lymphoid hyperplasia, as well as increased mast cell infiltration in the submucosa. In agreement with these results, O'sullivan et al.³⁷ and Öhman et al.³⁸ reported the presence of intestinal inflammation with an increase in T-lymphocytes and mast cells in IBS patients.

Treatment of rats with STW 5 tended to protect against these changes. At the lower dose level of 1 ml/Kg the changes were reduced to a measurable extent, but full protection was afforded by the higher dose (2.5 ml/Kg). In line with the present findings, it was reported that STW 5 in similar doses prevented the colonic histopathological inflammatory changes in a rat model of colitis³⁹ and reduced the severity of the inflammatory changes and esophageal lesions in a rat model of reflux esophagitis.²⁸

In parallel to the observed changes in colonic histopathology and inflammation transmitter levels, it was found that exposure of the rats to RS led to a decrease in the rate of weight gain as compared to

the control ones over the experimental period. Treatment with STW 5 restored the normal rate of weight gain. In other inflammatory conditions of the colon, such as ulcerative colitis, rats were shown to exhibit marked reduction in the rate of body weight gain during induction of colitis, an effect that was normalized by STW 5.³⁹

5 | CONCLUSION

The present findings have shown that exposure of rats to RS has led to changes in serotonin levels along the brain-gut axis with related changes in plasma CRF. This, in turn, has led to a disturbance in the level of mucosal mast cell inflammatory mediators leading to the development of IBS-like symptoms. The effectiveness of STW 5 in the latter condition may be related, at least in part, to its ability to protect against the above changes.

AUTHOR CONTRIBUTION TO THE STUDY

NGF, RMA, OAA, LNG contributed to the analysis and interpretation of data and drafting of the manuscript, while MTK contributed toward the design and guidance throughout the study as well as critical revision of the manuscript.

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CONFLICT OF INTEREST

Laila N. Gadelrub is fully employed at EVA Pharma, Cairo, Egypt. All other authors have no competing interests.

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